INCH-POUND

MIL-C-82811(OS)
29 December 1989
SUPERSEDING
WS 22166A
15 July 1986

MILITARY SPECIFICATION

COMPOUND, MOLD RELEASE, LIQUID

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 **Scope.** This specification covers the requirements for one type of liquid mold release compound, referred to herein as the release compound.

2. APPLICABLE DOCUMENTS

2.1 **Government documents.**

2.1.1 **Specifications, standards and handbooks.** The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

STANDARDS

FEDERAL

FED-STD-313 Material Safety Data, Transportation Data and Disposal Data for Hazardous Materials Furnished to Government Activities

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Ordnance Station, Standardization Branch (3730), Indian Head, MD 20640-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8030

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MILITARY

MIL-STD-129 Marking for Shipment and Storage

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from: Military Specifications and Standards, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094).

2.1.2 **Other Government documents, drawings, and publications.** The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

CODE OF FEDERAL REGULATIONS (CFR)

49 CFR 100-199 Transportation

(Application for copies of CFRs should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402-0001.)

2.2 **Non-Government publications.** The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D 542 Index of Refraction of Transparent Organic Plastics (DoD adopted)

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187.)

AMERICAN TRUCKING ASSOCIATION, INC.

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Association, Inc., Attn: Traffic Dept., 2200 Mill Road, Alexandria, VA 22314-4677.)

NATIONAL RAILROAD FREIGHT COMMITTEE

Uniform Freight Classification (UFC) 6000

(Application for copies should be addressed to the National Railroad Freight Committee, 222 South Riverside Plaza, Suite 1120, Chicago, IL 60606-5945.)

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(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents may also be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample of the release compound shall be subjected to first article inspection (see 6.4) in accordance with 4.3.

3.2 Material. The liquid mold release compound shall be an off-white aqueous solution containing small white flocculent particles.

3.3 Physical and chemical properties. Physical and chemical properties of the release compound shall conform to table I.

TABLE I. Physical and chemical properties.

Property	Limits	
	Minimum	Maximum
Hydrogen ion (pH)	7.7	9.2
Refractive index	1.343	1.357
Solids content, percent	5.5	8.5

3.4 Shelf life. The release compound shall have a shelf life of 12 months after date of acceptance when stored at ambient temperature in the original closed containers.

3.4.1 Shelf life extension. The shelf life expiration date of an individual lot may be extended for additional 12-month periods provided the release compound, upon retest, conforms to the solids content requirement of 3.3.

3.5 Safety.

3.5.1 Toxic products. The material shall have no adverse effect on the health of personnel when used for its intended purpose. Questions pertinent to this effect shall be referred by the contracting activity to the appropriate departmental medical service who will act as an advisor to the contracting agency (see 4.1.2).

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3.5.2 **Material Safety Data Sheets (MSDS).** The contractor shall prepare and submit an MSDS in accordance with FED-STD-313 as specified in the contract (see 6.2).

3.6 **Workmanship.** Workmanship shall be such that the release compound is uniform in appearance and free from visible contamination.

4. QUALITY ASSURANCE PROVISIONS

4.1 **Responsibility for inspection.** Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 **Responsibility for compliance.** All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 **Toxicological product formulations.** The contractor shall have the toxicological product formulations and associated information available for review by the contracting activity to evaluate the safety of the material for the proposed use.

4.2 **Classification of inspections.** The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).

b. Quality conformance inspection (see 4.4).

4.3 **First article inspection.** First article inspection shall consist of the quality conformance inspections (see 4.4). Failure to meet all of the requirements specified herein shall be cause for rejection of the first article sample.

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4.4 **Quality conformance inspection.** Quality conformance inspection shall consist of the examination and tests specified in table II.

TABLE II. Quality conformance inspection.

Examinations and Tests	Requirement	Test Method
Visual examination	3.2, 3.6, and Section 5	4.5.1
pH	3.3	4.5.2
Refractive index	3.3	4.5.3
Solids content	3.3	4.5.4

4.4.1 **Lot.** A lot shall consist of all the release agent manufactured by identical processes from one of the following:

a. A single batch (see 6.3.1).

b. A continuous process (see 6.3.2).

c. A uniform blend of batches or a uniform blend of portions from a continuous process.

4.4.2 **Sampling.** Random samples shall be taken from each lot as specified in table III.

TABLE III. Sampling.

No. of containers in the Lot	No. of containers to be Sampled
1 - 3	All
4 - 40	3
41 - 65	4

4.5 **Inspection methods.** The following inspection methods shall be used. Unless otherwise specified in the applicable test method, all weights, volumes, temperatures, and times shall be measured to the nearest specified unit or decimal.

NOTE: Reagent grade chemicals shall be used for chemical reactions in the conduct of all tests defined herein. Solvents and indicators may be commercial nonreagent grade materials.

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4.5.1 Visual examination. Visual examination shall be made to verify conformance to the requirements of 3.2, 36, and section 5.

4.5.2 pH. pH shall be determined using a buffer-standardized pH meter in accordance with the instructions supplied with the instrument and shall be recorded to the nearest 0.01 unit.

4.5.3 Refractive index. Refractive index shall be determined in accordance with ASTM D 542 and shall be recorded to the nearest 0.002 unit.

4.5.4 Solids content. Solids content shall be determined in accordance with the following:

a. Weigh a sample (A) of 10 ± 2 grams (g) to the nearest milligram (mg) into a tared, flat aluminum weighing dish. Do not agitate container prior to obtaining sample.

b. Place the dish with sample into a forced draft oven for a minimum of 3 hours at 225 ± 25°F.

c. Remove the dish with sample from the oven and place in a desiccator to cool.

d. When cooled to room temperature, weigh to the nearest mg and replace in the oven for an additional 15 minutes minimum.

e. Remove from the oven and cool in a desiccator to room temperature.

f. When cooled to room temperature, remove from the desiccator and weigh to the nearest mg (B).

g. Repeat b through f until constant weight is obtained.

h. Calculate the percent solids as follows:

$$\text{Percent solids} = \frac{B}{A} \times 100$$

i. Record the percent solids to the nearest 0.01 percent.

4.5.5 Storage life extension. When required, the tests for storage life extension specified in 3.4.1 shall be in accordance with 4.5.4.

4.6 Inspection of packaging. The release compound containers and container markings shall be visually examined to verify compliance with Section 5.

5. PACKAGING

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5.1 **Packaging.** Unless otherwise specified in the contract or order (see 6.2), packaging shall be Level C as specified herein.

5.1.1 **Level C.** The release compound shall be packaged to afford adequate protection against loss, contamination, deterioration, and damage during shipment from the supply source to the first receiving activity and during storage under the shelf life period and conditions specified in 3.4. Containers in the same shipment shall be of the same size. The packaging shall conform to the UFC 6000, National Motor Freight Classification, 49 CFR 171-178, or to other carrier rules and regulations as applicable to the mode of transportation.

5.2 **Marking.**

5.2.1 **Standard marking.** In addition to any special marking required by the contract or order (see 6.2), interior and exterior containers shall be marked in accordance with MIL-STD-129.

5.2.2 **Special marking.** In addition to the marking requirements of 5.2.1, each container marking shall include the following information:

a. Supplier identification, lot number, and material name.

b. Date of manufacture.

c. Net weight of contents.

d. Storage conditions.

e. Shelf life and inspection/test date for extension (see 3.4.1). In addition, the marking shall include the following:

"SHELF LIFE EXTENSION

THE SHELF LIFE OF AN INDIVIDUAL LOT MAY BE EXTENDED FOR AN ADDITIONAL 12 MONTHS PROVIDED THE RELEASE COMPOUND, UPON RETEST, CONFORMS TO THE PERCENT SOLIDS REQUIREMENTS OF MIL-C-82811."

f. Toxicity warning (if applicable).

g. Mark each container with supplier identification, vendor lot number, material name, acceptance stock and lot number, and date of acceptance testing.

6. **NOTES**

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

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6.1 **Intended use.** The release compound is intended for use in the manufacture of the Standard Missile Mk 104 Dual Thrust Rocket Motor nozzle components.

6.2 **Acquisition requirements.** Acquisition documents must specify the following:

a. Title, number and date of this specification.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.1.2).

c. Whether first article inspection is required (see 3.1)

d. Inspection and test data required (see 4.3)

e. Material Safety Data Sheets required (see 3.5.2 and 6.5)

f. Size of container, if applicable (see 5.1 and 5.2)

g. Special marking, if other than as specified in 5.2.2

h. Safety precautions required (see 6.6)

i. Acceptance stock and lot number

6.3 **Definitions.**

6.3.1 **Batch.** A batch is that amount of curing agent made as one unit in the manufacturing process.

6.3.2 **Continuous process.** A continuous process is an uninterrupted process of manufacture from which, in one unbroken flow, the curing agent is accumulated for the lot.

6.4 **First article.** When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item(s) should be a preproduction sample, a first article sample, a first production item, a sample selected from the first production items, or a standard production item from the contractor's current inventory (see 3.1), and the number of items to be tested as specified in 4.4. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending

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contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.5 **Material Safety Data Sheets.** MSDS requirements are applicable to this specification and should be specified in the contract as required by the Federal Acquisition Regulation (FAR) Part 23, subpart 23.3. Contracting officers will identify those activities requiring copies of completed Material Safety Data Sheets prepared in accordance with FED-STD-313. The pertinent Government mailing addresses for submission of data are listed in FED-STD-313.

6.6 **Safety precautions.** The safety precaution requirements of the "Contractor's Safety Manual for Ammunition, Explosives, and Related Dangerous Material" (DOD 4145.26M) are applicable and should be specified in the contract as required by the DOD FAR Supplement 28.7102.

NOTE: When this specification is used as part of the description of work to be accomplished by a Government activity, the safety precaution requirements of "Ammunition and Explosives Ashore" (OP5) should be made applicable.

6.7 **Subject term (key word) listing.**

Rocket Motor, Dual Thrust, Mk 104
Standard Missile

Preparing Activity:
NAVY-OS
(Project 8030-N112)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-C-82811	2. DOCUMENT DATE (YYMMDD) 29 December 1989

3. DOCUMENT TITLE
COMPOUND, MOLD RELEASE, LIQUID

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY NAVAL ORDNANCE STATION (CODE 3730) INDIAN HEAD, MD 20640-5000

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON
c. ADDRESS (Include Zip Code)	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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